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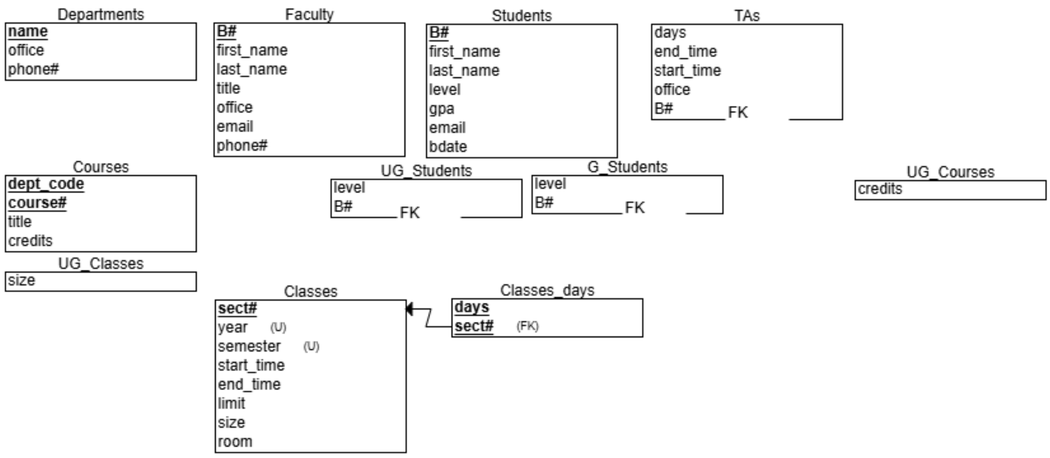
Homework 2

INFO 532-01

"I have done this assignment completely on my own. I have not copied it, nor have I given my solution to anyone else. I understand that if I am involved in plagiarism or cheating, I will have to sign an official form that I have cheated and that this form will be stored in my official university record. I also understand that I will receive a grade of 0 for the involved assignment and my grade will be reduced by one level (e.g., from A to A- or from B+ to B) for my first offence, and that I will receive a grade of "F" for the course for any additional offence of any kind."

Sign: Weihua Zheng

1. ER->Relational



Department

Candidate keys: phone#, office (both unique)

Faculty

Candidate keys: email (unique)

Foreign keys: None

Constraints:

- title must be one of {adjunct, lecturer, assistant professor, associate professor, professor}
- B# follows "B" + 8 digits
- Multiple faculty members may share office and phone#

Students

Candidate keys: email (unique)

TAs

Candidate keys: email (unique)

Foreign keys: B# → Student(B#)

Constraints:

- level must be {master, PhD}

- office_hours format: (start_time, end_time, days)

Courses

Constraints:

- cid follows format {dept_code + 3-digit course#}
- Undergraduate course#: 100-499 (4 credits), Graduate course#: 500-799 (3 credits)

Classes

Foreign keys: cid → Course(cid)

Constraints:

- semester is one of {Spring, Fall, Summer 1, Summer 2, Winter}
- size ≤ limit
- Undergraduate class must have at least 10 students; Graduate class must have at least 6 students
- start_time < end_time

Additional Constraints:

- Class sizes cannot exceed their limit.
- A student must complete all prerequisites with at least a C before enrolling in a course.
- A student's class schedule cannot have overlapping times.
- No faculty member can be assigned to overlapping classes.
- A student must enroll in at least one class per semester.

2. For example1, one class can have multiply TA, but one TA can only assist one class. This can be represented by min-max card in the E-R, but it cannot be explicitly represented in the relational data.
   Example2, TAs Must Be Graduate Students, ER model can use IS_A hierarchy to represent, but relational model is hard to represent
   Example3, prerequisite class, in ER, can be present in cycle a prerequisite. but relational model does not provide a direct way to enforce the constrain.

3. The maximum number of possible keys for a relation R(A, B, C) is 7.
   {A}: If A alone can uniquely identify a tuple, then {A} is a candidate key.
   {B}: If B alone can uniquely identify a tuple, then {B} is a candidate key.
   {C}: If C alone can uniquely identify a tuple, then {C} is a candidate key.
   {A,B}: If neither A nor B alone can uniquely identify a tuple, but together they can, then {A,B} is a candidate key.
   {A,C}: If neither A nor C alone can uniquely identify a tuple, but together they can, then {A,C} is a candidate key.
   {B,C}: If neither B nor C alone can uniquely identify a tuple, but together they can, then {{B,C} is a candidate key.
   {A,B,C}: If no subset of {A,B,C} can uniquely identify a tuple, then {A,B,C} is a candidate key.

4. The Entity Integrity Rule states that:
   - Every table must have a primary key.

- The primary key attribute(s) cannot contain null values.
- The primary key must uniquely identify each tuple in the relation。

The Referential Integrity Rule states that:

- If a foreign key in one table references a primary key in another table, then:
- Every value of the foreign key must either match a value of the referenced primary key or be null.
- The referenced primary key must exist in the referenced table

If we insert a tuple into R1, we must give a value, because primary key cannot be null. also A is a foreign key referencing B, this means the value of A in R2 already exist. But R2 is empty initially, which violate referential integrity rule. If we insert a tuple into R2, verse versa, the same situation.